

The adrenal glands are divided into two anatomically distinct regions: the adrenal medulla and the adrenal cortex. The adrenal medulla is a collection of irregularly shaped cells located in the interior of each adrenal gland. In some instances, increased serum concentrations of corticosteroid hormones can stimulate medullary cells to synthesize and secrete the catecholamines norepinephrine and epinephrine. Norepinephrine and epinephrine are water-soluble compounds that can function as both hormones and as neurotransmitters, and their secretion into the bloodstream is stimulated by acetylcholine (ACh)-mediated signaling in the sympathetic component of the autonomic nervous system. Figure 1 shows the enzyme-mediated synthesis of norepinephrine and epinephrine.

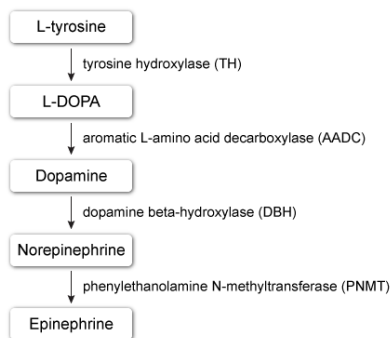


Figure 1

The adrenal cortex is the outer portion of the adrenal gland and is composed of cells that synthesize and secrete lipophilic hormones derived from cholesterol. Hormones produced by the adrenal cortex include the principal glucocorticoid hormone (cortisol) and the principal mineralocorticoid hormone (aldosterone).

The hypothalamus and pituitary gland regulate and control the endocrine functions of the adrenal cortex (Figure 2). Secretion of corticotropin-releasing hormone (CRH) from the hypothalamus promotes the synthesis and release of adrenocorticotrophic hormone (ACTH) by the pituitary gland. ACTH acts on adrenal cortex cells to promote the growth of cortical tissue and to increase the synthesis and secretion of glucocorticoids. In addition to its body-wide functions as the principal stress hormone, cortisol also regulates adrenal activity by inhibiting CRH release from the hypothalamus and ACTH release from the pituitary.

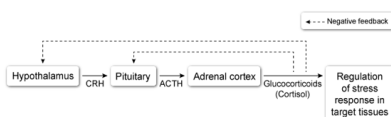


Figure 2

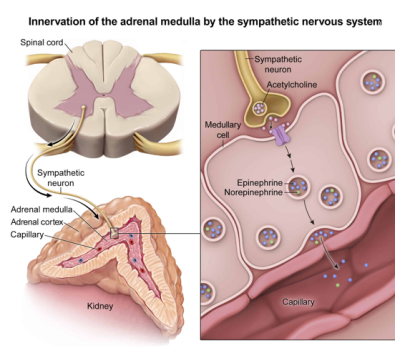
Administration of a drug causes increased secretion of norepinephrine and epinephrine by the adrenal medulla. This likely occurs because the drug:

- ☐ A. decreases secretion of ACTH. [4%]
- ☐ B. inhibits production of TH and AADC. [2%]
- ☒ C. mimics the activity of sympathetic neurons. [76%]
- ☐ D. increases transport of dopamine out of medullary cells. [17%]

Correct

75% answered correctly

Explanation:



The **autonomic nervous system** refers to the collection of neurons and peripheral nerves that regulate the involuntary functions of smooth muscles, cardiac muscles, and endocrine organs. The **autonomic nervous system** is composed of **two branches** with

broadly **antagonistic effects**: the **sympathetic nervous system** and the **parasympathetic nervous system**. Many tissues innervated by the autonomic nervous system are innervated by both branches.

Innervation of the adrenal medulla by the autonomic nervous system is unique in that the adrenal medulla is *innervated only by the sympathetic nervous system*. The passage states that norepinephrine and epinephrine release is stimulated by ACh-mediated signaling in the sympathetic component of the autonomic nervous system. Accordingly, for a drug to trigger an increased secretion of norepinephrine and epinephrine by the adrenal medulla, the **drug must** generate effects that **mimic the activity of the sympathetic nervous system**.

(Choice A) Figure 2 in the passage shows that ACTH is released by the pituitary gland. ACTH release generally stimulates secretion of glucocorticoids and mineralocorticoids from the adrenal cortex, and the passage states that in some circumstances, glucocorticoids can stimulate synthesis and secretion of epinephrine and norepinephrine. Therefore, a drug-induced increase (*not* decrease) in ACTH release could cause increased secretion of norepinephrine and epinephrine.

(Choice B) Figure 1 shows that the enzymes TH and AADC catalyze the precursors to norepinephrine and epinephrine. Consequently, if these enzymes were inhibited, norepinephrine and epinephrine levels would decrease (*not* increase).

(Choice D) According to Figure 1, norepinephrine and epinephrine are derived from the amino acid L-tyrosine, with formation of dopamine as an intermediate step. Therefore, increasing transport of dopamine out of the cell would likely decrease (*not* increase, as was observed with drug administration) secretion of epinephrine and norepinephrine.

Educational objective:

The autonomic nervous system is divided into the parasympathetic and sympathetic divisions, which are generally antagonistic in function. Although most tissues are innervated by both sympathetic and parasympathetic motor fibers, the adrenal medulla is unique in that it is innervated only by the sympathetic nervous system.

Time spent:0 sec

Subject:
Biology

QID:401706

System:
Endocrine and Nervous Systems

The adrenal glands are divided into two anatomically distinct regions: the adrenal medulla and the adrenal cortex. The adrenal medulla is a collection of irregularly shaped cells located in the interior of each adrenal gland. In some instances, increased serum concentrations of corticosteroid hormones can stimulate medullary cells to synthesize and secrete the catecholamines norepinephrine and epinephrine. Norepinephrine and epinephrine are water-soluble compounds that can function as both hormones and as neurotransmitters, and their secretion into the bloodstream is stimulated by acetylcholine (ACh)-mediated signaling in the sympathetic component of the autonomic nervous system. Figure 1 shows the enzyme-mediated synthesis of norepinephrine and epinephrine.

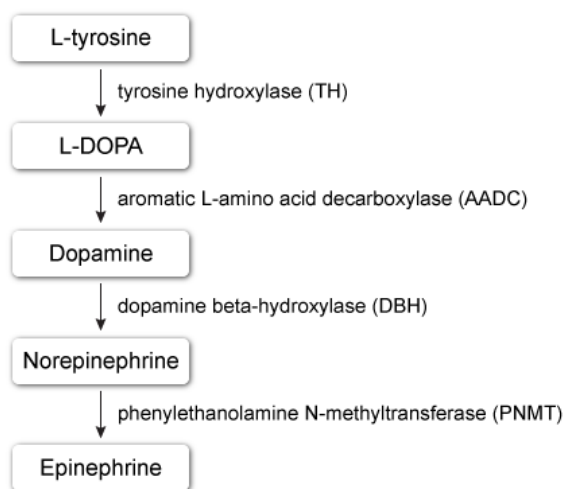


Figure 1

The adrenal cortex is the outer portion of the adrenal gland and is composed of cells that synthesize and secrete lipophilic hormones derived from cholesterol. Hormones produced by the adrenal cortex include the principal glucocorticoid hormone (cortisol) and the principal mineralocorticoid hormone (aldosterone).

The hypothalamus and pituitary gland regulate and control the endocrine functions of the adrenal cortex (Figure 2). Secretion of corticotropin-releasing hormone (CRH) from the hypothalamus promotes the synthesis and release of adrenocorticotrophic hormone (ACTH) by the pituitary gland. ACTH acts on adrenal cortex cells to promote the growth of cortical tissue and to increase the synthesis and secretion of glucocorticoids. In addition to its body-wide functions as the principal stress hormone, cortisol also regulates adrenal activity by inhibiting CRH release from the hypothalamus and ACTH release from the pituitary.

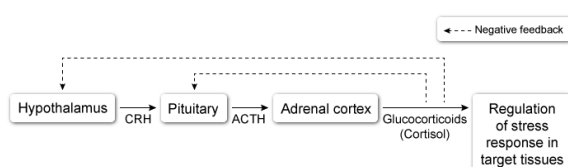


Figure 2

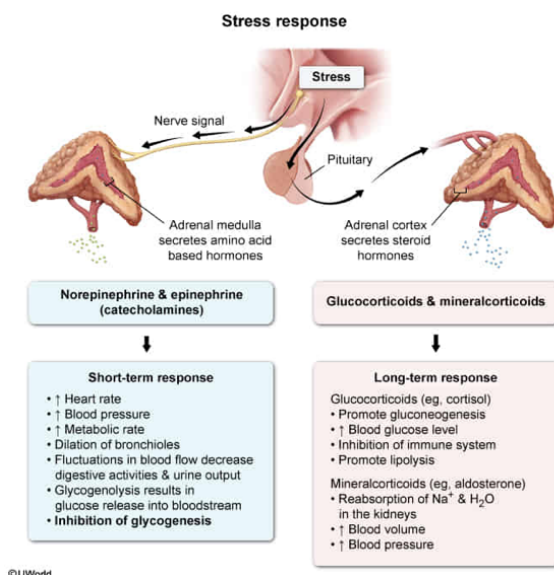
Hormones secreted by the adrenal medulla most likely influence energy metabolism by:

- ✓ ☒ A. inhibiting proteins that mediate glycogen synthesis. [54%]
- ☐ B. activating their cytosolic receptors in target cells. [21%]
- ☐ C. reducing the activity of enzymes that catalyze fat hydrolysis. [7%]
- ☐ D. increasing the activity of proteins that inhibit gluconeogenesis. [16%]

Correct

54% answered correctly

Explanation:



Energy metabolism refers to the cellular processes by which an organism stores or utilizes sources of energy. During periods of stress, hormones secreted by the adrenal gland influence energy metabolism by *promoting energy utilization* and *inhibiting energy storage*.

Hormones secreted by the adrenal gland regulate energy metabolism as follows:

- **Glucocorticoids** (eg, cortisol) released by the adrenal cortex act on the **liver** to stimulate the synthesis of glucose from other molecules (ie, **gluconeogenesis**) and the breakdown of fats into fatty acids (ie, lipolysis). Both glucose and free fatty acids are forms of cellular energy that can be readily utilized.
- **Norepinephrine** and **epinephrine** are released by the adrenal medulla and modulate the metabolism of glycogen, the storage form of glucose. Both hormones promote the breakdown of glycogen into glucose monomers in a process known as **glycogenolysis**. **By inhibiting the enzymes that mediate glycogen synthesis**, norepinephrine and epinephrine also *inhibit* **glycogenesis** (ie, the formation of glycogen from glucose).

(Choice B) *Water-insoluble* hormones (eg, steroid hormones) can diffuse across the plasma membranes of cells to bind **cytosolic receptors**. However, as stated in the

passage, the adrenal medulla secretes the *water-soluble* hormones norepinephrine and epinephrine. Water-soluble hormones *cannot* diffuse across the plasma membrane, and instead influence cellular activity by binding **extracellular surface receptors** to initiate a second messenger cascade.

(Choice C) For **triglycerides** (ie, the storage form of fatty acids) to be **utilized** as a source of energy, the fatty acid chains must be released from their glycerol backbone in a reaction facilitated **by lipases**. Because medullary hormones promote *energy utilization*, they would be expected to *increase* lipase activity, thereby *increasing* the use of fats as a source of energy.

(Choice D) Increasing the activity of proteins that *inhibit* gluconeogenesis would *decrease* glucose synthesis and *decrease* available energy. In addition, adrenal gland hormones *promote* gluconeogenesis such that the serum level of glucose rises.

Educational objective:

Energy metabolism refers to the processes by which the body manages cellular energy stores. The adrenal gland increases the body-wide level of free cellular energy sources by releasing glucocorticoids, norepinephrine, and epinephrine.

Time spent:0 sec

Subject:
Biology

QID:401707

System:
Endocrine and Nervous Systems

The adrenal glands are divided into two anatomically distinct regions: the adrenal medulla and the adrenal cortex. The adrenal medulla is a collection of irregularly shaped cells located in the interior of each adrenal gland. In some instances, increased serum concentrations of corticosteroid hormones can stimulate medullary cells to synthesize and secrete the catecholamines norepinephrine and epinephrine. Norepinephrine and epinephrine are water-soluble compounds that can function as both hormones and as neurotransmitters, and their secretion into the bloodstream is stimulated by acetylcholine (ACh)-mediated signaling in the sympathetic component of the autonomic nervous system. Figure 1 shows the enzyme-mediated synthesis of norepinephrine and epinephrine.

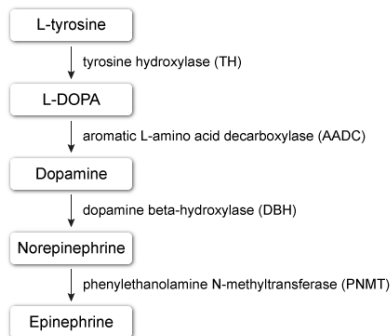


Figure 1

The adrenal cortex is the outer portion of the adrenal gland and is composed of cells that synthesize and secrete lipophilic hormones derived from cholesterol. Hormones produced by the adrenal cortex include the principal glucocorticoid hormone (cortisol) and the principal mineralocorticoid hormone (aldosterone).

The hypothalamus and pituitary gland regulate and control the endocrine functions of the adrenal cortex (Figure 2). Secretion of corticotropin-releasing hormone (CRH) from the hypothalamus promotes the synthesis and release of adrenocorticotropic hormone (ACTH) by the pituitary gland. ACTH acts on adrenal cortex cells to promote the growth of cortical tissue and to increase the synthesis and secretion of glucocorticoids. In addition to its body-wide functions as the principal stress hormone, cortisol also regulates adrenal activity by inhibiting CRH release from the hypothalamus and ACTH release from the pituitary.

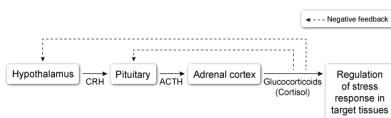


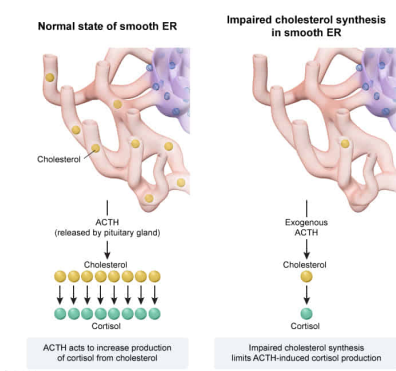
Figure 2

For individuals who exhibit insufficient hormone production by the adrenal cortex, ACTH is sometimes administered as a treatment option. Clinicians have reported that most patients fail to respond to this treatment with ACTH. A possible reason for this outcome is that nonresponsive patients:

- ☐ A. release insufficient amounts of ACTH from the pituitary gland. [8%]
- ☐ B. continuously overexpress the genes for DBH and PNMT. [4%]
- ☒ C. exhibit impaired cholesterol synthesis in the smooth endoplasmic reticulum of adrenocortical cells. [59%]
- ☐ D. express an overactive ACTH receptor on cells responsive to cortisol. [27%]

Correct
59% answered correctly

Explanation:



The **endoplasmic reticulum**, a cytosolic organelle composed of a network of membranous tubules, is located near the nucleus and is present in all human cells except red blood cells and spermatozoa. The endoplasmic reticulum can be divided into **two components**: the **rough endoplasmic reticulum (RER)** and the **smooth endoplasmic reticulum (SER)**.

The RER has ribosomes located along its surface that **translate the mRNA** of proteins destined for the **secretory pathway**. The SER contains no ribosomes along its surface; instead, the interior compartment (lumen) of the SER contains enzymes that synthesize **lipids**, including cholesterol and cholesterol-derived molecules (eg, **steroid hormones**).

ACTH is released by the anterior pituitary and acts on the adrenal cortex to *promote* the synthesis and secretion of cortisol, a steroid hormone produced from cholesterol. However, the *actual synthesis* of this steroid hormone depends on the availability of cholesterol from transport into the cell and from production of cholesterol by SER proteins, *not* ACTH. Consequently, defective SER proteins can compromise cortisol synthesis within the adrenal cortex regardless of ACTH stimulation. Accordingly, patients with **adrenal insufficiency** caused by **defective SER proteins** can exhibit a limited response to **ACTH administration**.

(Choice A) Insufficient ACTH secretion by the pituitary gland would result in adrenal insufficiency that *is* responsive to the administration of ACTH. This deficit is because *exogenous* ACTH stimulates the adrenal cortex to synthesize and secrete *glucocorticoid hormones*, correcting the adrenal insufficiency initially caused by reduced endogenous ACTH signaling.

(Choice B) Overexpression of the genes DBH and PNMT would lead to increased synthesis of norepinephrine and epinephrine by the adrenal medulla. This increase would cause adrenal *medulla* hyperactivity, a condition requiring treatments separate from those used in cases of adrenal insufficiency (eg, adrenal *cortex* hypoactivity).

(Choice D) A constitutively active ACTH receptor would mimic the effects of high serum ACTH levels, stimulating excessive glucocorticoid secretion by the adrenal cortex. This excess would result in adrenal cortex *hyperactivity*, not adrenal insufficiency (eg, adrenal cortex *hypoactivity*). ACTH administration, which promotes adrenal cortex function, would *not* be used as a treatment for these patients.

Educational objective:

The endoplasmic reticulum is a cellular organelle with rough and smooth components. Synthesis of lipids and steroid hormones takes place within the smooth endoplasmic reticulum.

Time spent: 0 sec
Subject:
Biology

QID: 401708
System:
Endocrine and Nervous Systems